

FITBUDDY

AI-Powered Personalized Workout and Wellness Assistant

PHASE 01

Brainstorming & Ideation Phase

Define the problem, target users, solution idea, scope, and value proposition for FitBuddy.

Prepared for phase-wise submission

30 September 2026

Phase objective: Define the problem, target users, solution idea, scope, and value proposition for FitBuddy.

1. Problem Statement

- Many people begin fitness routines without a structured weekly plan that matches their goal, preferred intensity, and current profile.
- A useful digital assistant should make workout planning simple, repeatable, and interactive while avoiding the impression that generated guidance is a substitute for professional medical advice.
- FitBuddy addresses this by combining a web form, a backend API, a database, and Generative AI services to produce a personalized seven-day wellness plan.

2. Ideation

- Core idea: create an AI fitness companion that turns a small set of user inputs into an actionable 7-day workout and wellness plan.
- Personalization dimensions: age, weight, fitness goal, and workout intensity.
- Interaction model: generate plan -> review plan -> give feedback -> revise plan.
- Supporting features: nutrition, hydration, recovery suggestions and a local user dashboard.

3. Proposed Solution

- FitBuddy is a browser-based application backed by FastAPI. User data is validated with Pydantic, persisted with SQLAlchemy/SQLite, and passed to AI generation services.
- The workout generator returns a structured seven-day plan. A fast AI service returns nutrition/recovery guidance. A feedback service revises the original plan while preserving the seven-day structure.
- A mock AI mode is included so the project can be tested locally without a Gemini API key.

4. Target Users & Stakeholders

- Primary users: individuals who want a simple personalized workout schedule.
- Project stakeholders: student/development team, evaluator or instructor, and local demonstration users.
- Development stakeholders: backend, frontend, database, and AI integration responsibilities can be divided by module.

5. Scope

- In scope: workout generation, nutrition/recovery guidance, feedback-based revision, SQLite storage, web UI, API docs, mock mode, and automated tests.
- Out of scope: medical diagnosis, treatment, prescription plans, advanced authentication, public cloud deployment, and wearable-device integration.

6. Expected Outcomes

- A working FitBuddy web application.
- A repeatable workflow for generating and revising a 7-day plan.
- A documented technical architecture and testable API.
- A demo-ready system with clear setup and execution instructions.

Ideation Decisions

Area	Decision
Product concept	AI-powered personalized workout and wellness assistant

Area	Decision
Primary output	Exactly 7 workout days plus nutrition/recovery tips
Personalization inputs	Username, user ID, age, weight, goal, intensity
Interaction	Generate, review, submit feedback, revise
Persistence	SQLite database with SQLAlchemy
AI approach	Structured JSON validated by Pydantic
Safety position	General wellness guidance; not medical advice

Phase Deliverables / Evidence

- Problem statement and project concept
- Scope and target users
- Solution concept and expected outcomes

Submission Note

This phase document is intended to be submitted together with the corresponding FitBuddy source project and evidence captured during execution. Keep API keys, personal credentials, and other secrets out of screenshots and submissions.